MySQL — Zero to Master

22 topics, grouped as Beginner → Intermediate → Advanced → Master. Work through the topics in order; type every code example yourself and finish the practice tasks.

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Beginner Level

01. Introduction to Databases and MySQL

A database stores data in an organised way. MySQL is a popular open-source relational database that stores data in tables (rows and columns) and is queried with SQL.

```
mysql -u root -p
SHOW DATABASES;
```

02. Installation and Basic Commands

Install MySQL Server and a client (CLI, Workbench, DBeaver). Create and select a database.

```
CREATE DATABASE school;
USE school;
```

03. Data Types and Creating Tables

`INT`, `BIGINT`, `DECIMAL`, `VARCHAR`, `TEXT`, `DATE`, `DATETIME`, `BOOLEAN`, `JSON`. Define keys and constraints.

```
CREATE TABLE students (
  id INT AUTO_INCREMENT PRIMARY KEY,
  name VARCHAR(100) NOT NULL,
  email VARCHAR(150) UNIQUE,
  dob DATE
);
```

04. INSERT, SELECT, UPDATE, DELETE (CRUD)

The four basic operations. Always use `WHERE` with `UPDATE` and `DELETE`.

```
INSERT INTO students (name, email) VALUES ('Arun', 'arun@x.com');
SELECT * FROM students WHERE id = 1;
UPDATE students SET name = 'Arun K' WHERE id = 1;
DELETE FROM students WHERE id = 1;
```

05. Filtering and Sorting

`WHERE`, `AND/OR/NOT`, `IN`, `BETWEEN`, `LIKE`, `IS NULL`, `ORDER BY`, `LIMIT`, `DISTINCT`.

```
SELECT name FROM students WHERE name LIKE 'A%' ORDER BY name LIMIT 10;
```

06. Constraints and Keys

`PRIMARY KEY`, `FOREIGN KEY`, `UNIQUE`, `NOT NULL`, `DEFAULT`, `CHECK`.

```sql
ALTER TABLE marks ADD FOREIGN KEY (student_id) REFERENCES students(id);
```

Intermediate Level

07. Aggregate Functions and GROUP BY

`COUNT`, `SUM`, `AVG`, `MIN`, `MAX` with `GROUP BY` and `HAVING`.

```
SELECT city, COUNT(*) AS total FROM students GROUP BY city HAVING total > 5;
```

08. Joins

`INNER`, `LEFT`, `RIGHT` joins combine related tables; understand cross and self joins.

```
SELECT s.name, m.score FROM students s
JOIN marks m ON m.student_id = s.id;
```

09. Subqueries and CTEs

Queries inside queries; Common Table Expressions (`WITH`) improve readability.

```
WITH top AS (SELECT student_id, AVG(score) a FROM marks GROUP BY student_id)
SELECT * FROM top WHERE a > 80;
```

10. Normalization and Schema Design

1NF, 2NF, 3NF remove duplication; model one-to-many and many-to-many with foreign keys and junction tables.

```
CREATE TABLE enrollments (student_id INT, course_id INT, PRIMARY KEY (student_id, course_id));
```

11. Built-in Functions

String, date, numeric, conditional (`CASE`, `IF`, `COALESCE`) functions.

```
SELECT CONCAT(first, ' ', last), DATEDIFF(NOW(), dob) FROM students;
```

12. Views and Altering Tables

Views save queries; `ALTER TABLE` changes structure.

```
CREATE VIEW top_students AS SELECT * FROM students WHERE gpa > 9;
```

Advanced Level

13. Indexes and Query Optimisation

Indexes speed up reads; use `EXPLAIN` to inspect plans; avoid functions on indexed columns and `SELECT *`.

```
CREATE INDEX idx_email ON students(email);
EXPLAIN SELECT * FROM students WHERE email = 'a@x.com';
```

14. Transactions and ACID

Group statements so all succeed or none do. InnoDB supports ACID and row-level locking; know isolation levels.

```
START TRANSACTION;
UPDATE acc SET bal = bal - 100 WHERE id = 1;
UPDATE acc SET bal = bal + 100 WHERE id = 2;
COMMIT;
```

15. Stored Procedures, Functions, Triggers

Server-side logic: procedures, functions, triggers and events.

```
DELIMITER //
CREATE PROCEDURE get_student(IN sid INT) BEGIN SELECT * FROM students WHERE id = sid; END //
```

16. Window Functions

`ROW_NUMBER`, `RANK`, `SUM() OVER`, `LAG` for analytics without collapsing rows.

```
SELECT name, score, RANK() OVER (ORDER BY score DESC) FROM marks;
```

17. JSON and Full-text Search

JSON columns with `->>`; full-text indexes for search.

```
SELECT data->>'$.city' FROM profiles;
```

18. Users, Privileges and Security

Create least-privilege users, avoid SQL injection with prepared statements, enable TLS, rotate passwords.

```
CREATE USER 'app'@'%' IDENTIFIED BY '...';
GRANT SELECT, INSERT ON shop.* TO 'app'@'%';
```

Master Level

19. Backup and Restore

Logical backups with `mysqldump`; binary logs for point-in-time recovery.

```
mysqldump -u root -p shop > shop.sql
mysql -u root -p shop < shop.sql
```

20. Replication and High Availability

Source-replica replication, read scaling, failover, Group Replication / InnoDB Cluster.

```
SHOW REPLICA STATUS\G
```

21. Partitioning and Scaling

Partition large tables, shard horizontally, use connection pooling and caching.

```
PARTITION BY RANGE (YEAR(created)) (...)
```

22. Real Project: Library / E-commerce Schema

Design tables, relations, indexes and reports (top sellers, monthly revenue) with sample data and stored procedures.

Interview Question Bank

Q1. DELETE vs TRUNCATE vs DROP?
DELETE removes rows (can filter, rollbackable); TRUNCATE empties the table fast; DROP removes the table itself.

Q2. INNER JOIN vs LEFT JOIN?
INNER returns matching rows; LEFT returns all left rows plus matches (NULL when none).

Q3. WHERE vs HAVING?
WHERE filters rows before grouping; HAVING filters groups after aggregation.

Q4. What is normalization?
Organising tables to reduce redundancy and anomalies (1NF-3NF).

Q5. What is an index?
A data structure (B-tree) that speeds up lookups at the cost of extra writes/storage.

Q6. What is ACID?
Atomicity, Consistency, Isolation, Durability - guarantees for transactions.

Q7. PRIMARY KEY vs UNIQUE?
Primary key identifies rows, is not null and only one per table; UNIQUE can be many and allows NULLs.

Q8. What is a foreign key?
A column referencing another table's key to enforce referential integrity.

Q9. How do you find slow queries?
Slow query log, `EXPLAIN`, Performance Schema.

Q10. What is SQL injection?
Attack that inserts SQL through input; prevented by prepared statements.

Capstone Project Ideas

1. Student management schema
2. Library database
3. E-commerce orders and reports
4. Hospital appointment system
5. Banking transactions